

Problem Set 3 — Triangles & Vectors

Physics 2026–27 • due at the start of class, Thursday, September 10

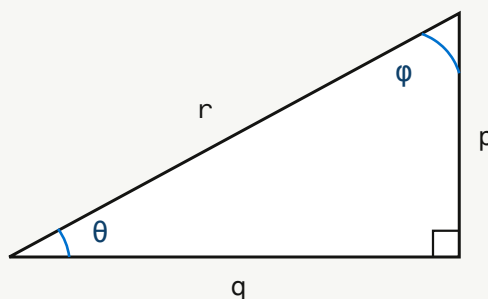
NAME _____

DATE _____

How to work this set. Every solved problem still runs the protocol: labeled **K / D / E / A**, units traveling with the math, symbols rearranged before numbers go in. The promise from class holds — **right triangles only**. You'll need a calculator with **sin**, **cos**, **tan** and their inverses. This is the last practice before the **Unit 0 Skills Check, Thursday, September 10** — the same morning this set is due.

1 Label the triangle

ANATOMY



- Which side is the hypotenuse? Give both reasons from class that identify it.
- Standing at angle θ : which side is opposite, and which is adjacent?
- Now stand at angle ϕ : which side is opposite, and which is adjacent?
- One sentence: why did two of your answers flip between (b) and (c) while one stayed put?

2 The zip line

SOH-CAH-TOA

A zip line cable runs **9.0 m** from a platform down to an anchor stake in the ground, making a **55°** angle with the ground at the stake. Run KDEA to find how high the platform is above the ground.

K	EVERY GIVEN, WITH UNITS
D	THE TARGET & ITS UNIT
E	WHICH RATIO? SOH • CAH • TOA
A	SYMBOLS FIRST • UNITS THE WHOLE WAY

3 Crossing the field

PYTHAGOREAN

A soccer player sprints 9 m down the sideline, then cuts 12 m straight across the field. Run KDEA to find the straight-line distance from where she started. No angle given, none needed — pick your tool accordingly.

Bonus: this triangle belongs to a famous family from class. Name the family, and write down the two non-right angles without touching your calculator.

K	EVERY GIVEN, WITH UNITS
D	THE TARGET & ITS UNIT
E	EQUATION
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

4 The ramp inspection

INVERSE TRIG

An access ramp rises 0.60 m over a horizontal run of 7.2 m. The building code says a ramp's angle with the ground may not exceed 5° . Run KDEA to find the ramp's angle, then rule: does it pass inspection?

K	EVERY GIVEN, WITH UNITS
D	THE TARGET & ITS UNIT
E	WHICH INVERSE UNDOES YOUR RATIO?
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

5 The kayaker

FULL VECTOR

A kayaker paddles 300 m east across a bay, then 400 m north along the shore. Find her displacement from the launch point — that means **both halves of the vector**:

- Magnitude:** the straight-line distance back to the launch.
- Direction:** the angle of that line, north of east.

K	EVERY GIVEN, WITH UNITS
D	TWO TARGETS, TWO UNITS
E	ONE TOOL PER TARGET
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

6 Splitting the climb

COMPONENTS

A small plane climbs a total distance of 6.0 km along a straight line angled 25° above the horizontal.

- How far did it travel horizontally? ($x = r \cos \theta$)
- How much altitude did it gain? ($y = r \sin \theta$)
- Check yourself:** put your two components back through $a^2 + b^2 = c^2$. What should come out, and does it?

K	EVERY GIVEN, WITH UNITS
D	THE TARGETS & THEIR UNITS
E	EQUATIONS
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

7 The loading dock

SOLVE FOR THE HYPOTENUSE

A delivery ramp must reach a dock 1.2 m above the parking lot, rising at 25° . Run KDEA to find how long the ramp itself has to be. Watch the rearrangement — this time the hypotenuse is the unknown, so the algebra divides where the zip line multiplied.

K	EVERY GIVEN, WITH UNITS
D	THE TARGET & ITS UNIT
E	WHICH RATIO?
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

8 Choose your tool

STRATEGY

Do **not** solve these. For each scenario, name the one tool you'd reach for — **SOH**, **CAH**, **TOA**, an *inverse* ratio, or the *Pythagorean theorem* — and one phrase saying how you knew.

- a. You know an angle and the hypotenuse; you want the side across from the angle.
- b. You know both legs; you want the hypotenuse, and no angle appears anywhere.
- c. You know both legs; you want the angle between the hypotenuse and one of them.
- d. You know a vector's magnitude and its angle above horizontal; you want the horizontal piece.

WORK

Graded on the homework 4–3–2–1 scale. Bring this set, your calculator, and everything from Workshops I–II with you Thursday — the Skills Check covers all three.